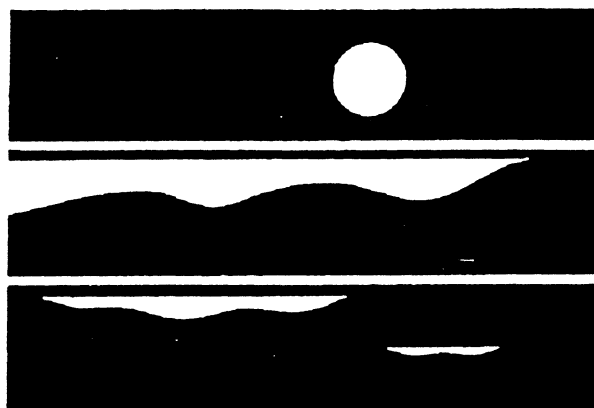




WASHINGTON STATE
DEPARTMENT OF
E C O L O G Y

Sulfur Dioxide Procedures

By
Barry Jespersen

Air Quality Program

July 1993

#95-201F

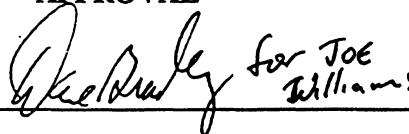
State of Washington
Department of Ecology
Air Quality Program

SULFUR DIOXIDE PROCEDURES

July 1993

APPROVAL

Joe Williams
Air Quality Program Manager

 for Joe Williams Date: 8/19/93

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 Date: 7/26/93

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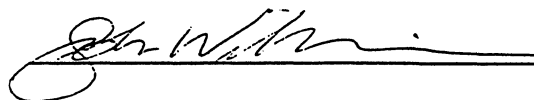
 Date: 7/9/93

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1.0 Introduction

This document describes the procedures involved in site selection, installation, calibration standards, calibration, operation, quality control, preventive and remedial maintenance, quality assurance, recording and validating data, and data quality assessment for sulfur dioxide (SO₂) ambient air monitoring. The procedures are applicable to air monitoring projects that generate SO₂ data used, supported or mandated by the Washington State Department of Ecology Air Quality Program.

These procedures are not instrument specific. For instrument specific operation refer to the manufacturer's manual.

2.0 Site Selection

Good air monitoring data is dependant upon several considerations. Of primary importance is the sampling network design and site selection.

Before selecting the monitoring site, determine the purpose of the monitoring from the list presented below:

- 1) To judge compliance with and/or progress made toward meeting ambient air quality standards.
- 2) To activate emergency control procedures that prevent or alleviate air pollution episodes.
- 3) To observe pollution trends throughout a region, including non-urban areas.
- 4) To provide a data base for research evaluation of effects: urban, land-use, and transportation planning; development and evaluation of abatement strategies; and development and validation of diffusion models.

Monitoring Considerations

The selection of an air monitoring site must take into account the following factors:

- spacial scales
- logistics
- economics
- pollutant considerations
- meteorological restraints
- topography
- representativeness

Once the monitoring purpose and constraints are identified, a search for an acceptable monitoring site can begin.

When searching for a monitoring site, identify several possible site locations. In this way the best location can be selected so as to ensure that the monitoring objectives and siting criteria are met.

For more detailed information concerning site selection refer to the Quality Assurance Handbook for Air Pollution Measurement Systems, Volume II, Section 2.0.1, entitled "Sampling Network Design and Site Selection."

For Ecology and local air authority operated sites see the Site Selection and Installation document.

Probe Siting Criteria

Once the station location has been identified, the following probe siting criteria must be used to ensure for the consistent collection of air monitoring data.

The most desirable height for an SO₂ monitoring inlet probe is near the breathing height. The inlet probe must be located 3 to 15 meters above ground level. If the inlet probe is located on the side of a building, then it should be located on the windward side of the building relative to the prevailing winter wind direction. The inlet probe must also be located more than 1 meter vertically or horizontally away from any supporting structure and also away from dirty dusty areas.

No furnace or incineration flues or other sources of SO₂ should be nearby. The separation distance is dependent on the height of the flues, type of waste or fuel burned, and the quality of the fuel. If the inlet probe is located on a roof or other structure, it must be at least 1 meter from walls, parapets, penthouses, etc.

The inlet probe must be located away from obstacles and buildings with the distance between being at least twice the height that the obstacle protrudes above the inlet probe.

Air flow must be unrestricted in an arc of at least 270° around the inlet probe and the predominant wind direction for the season of greatest pollutant concentration must be included in the 270° arc. If the probe is located on the side of building, 180° clearance is required.

Trees can provide surfaces for SO₂ absorption and act as an obstruction to normal wind flow patterns. To minimize the possible effects of trees on the measured SO₂ levels, the sampler should be placed at least 20 meters from the drip line of trees. However, in situations where trees could be classified as an obstruction, the sampler must be placed at least 10 meters from the drip line of the obstructing trees.

For quick reference table 2-1 summarizes the siting criteria for the placement of the SO₂ analyzer.

Table 2-1
SO₂ Probe Siting Criteria

Scale	Height Above Ground (meters)	Distance from Supporting Structure (meters)		Other Spacing Criteria
		Vertical	Horizontal ^a	
All	3-15	> 1	> 1	1. Should be > 20 meters from the dripline of trees and must be 10 meters from the dripline when the trees act as an obstruction.
				Distance from inlet probe to obstacle, such as buildings, must be at least twice the height the obstacle protrudes above the inlet probe. ^b
				Must have unrestricted airflow 270° around the inlet probe, or 180° if the probe is on the side of a building.
				No furnace or incinerator flues should be nearby. ^c

^a When probe is located on rooftop, this separation distance is in reference to walls, parapets, or penthouses located on the roof.

^b Sites not meeting this criterion would be classified as middle scale.

^c Distance is dependent on height of furnace or incineration flues, type of fuel or waste burned, and quality of fuel (sulfur, ash or lead content). This is to avoid undue influences from minor pollution sources.

The Monitoring Station

Every monitoring station is unique. Monitoring equipment may be housed in a trailer, a room in a building, or a shelter specifically designed for air monitoring purposes. The following considerations should be made:

- 1) Is the instrument location dry and clean? Is it heated and air conditioned? Automated instruments must be maintained in a dry, clean, temperature-controlled (20 to 30°C) environment.
- 2) Will the instrument location need cabinets, table or shelves?
- 3) Will the site require a stand-alone shelter?

- 4) Does the site need electricity or phone lines brought in and are contracts for construction needed.
- 5) Will the site be easily accessible by the operator? i.e. owners approval to access the site.
- 6) Is additional security necessary?

3.0 Site Installation

Once a site has been selected and the monitoring station has been established, equipment and supplies are required for the measurement of SO₂. The station must contain the following equipment;

- SO₂ analyzer and manual
- SO₂ calibrator
- permeation tube
- data acquisition system
- strip chart recorder
- thermometer
- barometer
- voltmeter
- probe material
- station logbook
- spare parts
- supplies and expendables

Install the SO₂ monitoring station equipment according to the manufacturers manual. The information contained in the manufacturers manual include instructions on:

- installation of the instrumentation
- calibration
- operation
- preventive maintenance schedule and procedures
- troubleshooting

The manufacturers manual must be kept with the analyzer at all times. Ensure that the analyzer and calibrator are properly vented to the outside atmosphere to avoid any shelter contamination.

Figure 3-1 is an example of a completed SO₂ Siting Criteria Checklist (see Section 12 for a blank form). This form is to be used when installing an SO₂ site to ensure that all siting criteria established in 40 CFR, Part 58 are met.

Figure 3-1
Sulfur Dioxide (SO₂)
Siting Criteria Checklist

Site Name: Roosevelt Middle SchoolSite #:

0	5	6	6	0	0	8	A
---	---	---	---	---	---	---	---

Address: 400 S. Monroe Rd.City: Port Angeles County: Clallam

1. Vertical distance from ground to probe (3-15 meters): (Probe height must be less than 80% of the mean height of buildings in the neighborhood. This applies whether probe is located on a rooftop or at an intermediate height on a building. The 80% figure applies only to surrounding structures not the building where the probe is located. Minimum height must still be greater than or equal to 3 meters. NOTE: Inlet probe must be located away from dirty, dusty areas. 10 Meters
2. Distance from support structure (> 1 meter): (This separation distance is in reference to walls, parapets or penthouses located on the roof.) Vertical 4 Meters
Horizontal 4 Meters
3. Unrestricted air flow in an arc of 270° or 180° if on side of building. Predominant wind direction must be included in 270° arc. (The inlet probe should be on the windward side of the building relative to the prevailing winter wind direction. At least one of the directions must be the dominant direction for the season of greatest potential pollutant concentration.) Yes ☒ No ☐
4. Distance from probe to trees (should be >20 meters). 21.6 Meters
5. Distance from sampler to any obstacle must be at least twice the height the obstacle protrudes above the sampler. 15.4 Meters
6. Are any furnace or incineration flues nearby? (The separation distance is dependent on the height of the flue, type of material burned and quality of fuel. Objective is to avoid undue influences from minor local sources.) Yes ☐ No ☒
7. Comments: _____

Completed by: Don Wright Date: 10/23/91Agency: Ecology

4.0 Calibration Standards

The sulfur dioxide standards must be certified utilizing EPA's "Traceability Protocol for Establishing True Concentrations of Gases Used for Calibrations and Audits of Air Pollution Analyzers (Protocol No. 2)" as specified in EPA's Quality Assurance Handbook for Air Pollution Measurement Systems, Volume II (EPA/600/4-77-027a), Section 2.0.7 as well as employing the procedures contained in 40 CFR 58, Appendix A, "Quality Assurance Requirements for State and Local Air Monitoring Stations."

5.0 Calibration

The accuracy and precision of data derived from the air monitoring equipment are dependent on the quality assurance procedures used, primarily the instrument calibration.

The following calibration procedure involves introducing gas samples of known concentrations into the instrument, adjustments are made and then a calibration relationship is derived.

A temperature controlled calibration system is used to generate low SO₂ concentrations. This is done by holding a permeation tube at a constant temperature (+/- 0.1°C) upon which the permeation rate is known. The SO₂ permeating from the tube is carried away by clean air to a mixing chamber where it is accurately diluted with zero air to the desired concentration.

The procedure for a multipoint calibration is not instrument specific. For specific analyzer calibration instructions refer to the manufacturer's manual.

Use the Air Monitor Calibration Report form contained in Section 12 to aid in the collection and documentation of the calibration data. A completed example of this form is shown in Figure 5-3.

- 1) Record the official name and address of the station on the form. The information should be the same as that appearing on the site identification form to eliminate any confusion by persons not familiar with the station.
- 2) Identify the analyzer being calibrated by the manufacturer's name, model, and serial number.
- 3) Identify the person performing the calibration and the date of calibration.
- 4) Identify the calibrator by the manufacturer's name, model, and serial number.
- 5) Identify the permeation tube by tube serial number.

- 6) Record the barometric pressure and the shelter temperature.
- 7) Record the analyzer sample air flow.
- 8) Record the zero and span settings after the calibration is completed. (These setting will be used as a basis of comparison when changes occur in the instrument performance.)
- 9) Record the temperature at which the permeation tube is maintained during calibration and use the recorded temperature to determine the SO₂ permeation rate (ng/min) from the permeation tube data form supplied by the manufacturer. The chamber temperature must be maintained at 30°C +/- 0.10°C. A 1.0°C error in the chamber temperature will cause a 7% error in the permeation rate.

Multipoint Calibration

Use the calibration equations (Figures 5-1 and 5-2) and the Air Monitoring Calibration Report (Section 12) to record the data derived during the calibration of the SO₂ analyzer. A completed calibration report is shown in Figure 5-3. For instrument specific instructions refer to the manufacturer's manual before performing the following calibration procedure.

- 1) Select the operating range of the analyzer to be calibrated. Normally this range is 0 - 0.5 PPM. Special conditions may allow for the use of 0 - 1.0 ppm or 0 - 2.0 ppm operating ranges.
- 2) Connect the voltage output cable(s) of the analyzer to the input terminals of the strip chart recorder(s) and data logger (positive output cable to positive input connection and negative output cable to negative input connection). Verify that the analyzer, strip chart and data logger are all indicating the same value (within +/- 1% of full scale). All Adjustments to the analyzer should be performed based on volt meter measurements obtained from the analyzer outputs.
- 3) Attach the analyzer sample line to the output manifold of the calibrator. Adjust the zero air flow from the calibrator so that it exceeds the total demand of the analyzer to ensure that no ambient air is pulled into the manifold vent. Verify the flow by using either the rotameter on the front of the analyzer or a portable flow meter. Always check and make sure that the chamber overflow vent is open when introducing gas directly into the analyzer. Reverify that the chamber temperature is at 30°C +/- 0.10°C
- 4) Allow the analyzer to sample zero air until a stable response is obtained (strip chart trace does not deviate more then 2% of full scale in 10 minutes), then

adjust the analyzer zero control if necessary. Confirm that the strip chart is operating by verifying the zero reading.

- 5) Calculate the 80% upper range limit (URL), calculate the flow rate needed to generate the URL ppm reading using the equation shown in Figure 5-1:

Figure 5-1

$$F = (Pr/PPM) \times (M_c/M_w)$$

F = Flow (cc/min)

Pr = Perm tube perm rate (ng/min)

PPM = Gas concentration desired (ppm)

M_c = Molar constant (22.4 l/mole at 30°C)

M_w = Mole weight of SO₂ (64 g/mole)

Adjust the dilution air flow until the total air flow (dilution flow + carrier flow) from the calibrator is approximately the flow calculated above.

- 6) Calculate the actual concentration of SO₂ (ppm) being generated using the following equation:

Figure 5-2

$$PPM(SO_2) = (Pr/F) \times (M_c/M_w)$$

- 7) Allow the analyzer to sample the calibration gas until the response is stable, normally about 10 to 15 minutes, and then adjust the analyzer span until the analyzer output is the same as the calculated response. Verify the strip chart response. If adjustment of the span is required, recheck the zero and span adjustments several times to get a stable response.
- 8) Without adjusting the analyzer zero and span settings, generate one additional point at approximately 20% of the operating range by using the equations in Figures 5-1 and 5-2, substituting the new percentage value into the equations and adjusting the total flow (F). Allow the point to stabilize (strip chart trace does not deviate more than 2% of full scale in 10 minutes). Do not readjust the analyzer zero and span settings once the calibration has been completed.
- 9) Upon completion of the calibration regenerate the zero, low span point and high span point. Record the indicated concentration (voltmeter reading converted to PPM reading) and actual concentration (calculated concentration) on the air monitor calibration report Figure 5-3).

A straight line of best fit (for all points including zero) can now be calculated by the method of least squares (see Appendix J of Volume I, Quality Assurance Handbook for Air Pollution Measurement Systems). Because the time required to manually perform the calculation is considerably long, a programmable calculator is suggested.

To ensure the accurate measurements of the SO₂ concentrations, the analyzer should be calibrated prior to installation and reverified at the time of installation. Additional calibrations should be performed after a performance audit has indicated the analyzer response to be unacceptable. Based on performance audits conducted as specified in 40 CFR 58, Appendix A, the individual instrument accuracy must be within +/- 10% to be considered acceptable. A calibration should also be performed following any one of the activities listed below:

- 1) an interruption of more than a few days in analyzer operation,
- 2) any repairs,
- 3) physical relocation of the analyzer, or
- 4) any other indication (including excessive zero or span drift) of possible significant inaccuracy of the analyzer.

6.0 Quality Control Checks

Essential to quality assurance are scheduled visits to the monitoring location to verify the operational status of the monitoring system. A weekly visit should be made to the monitoring site. These visits include performing precision checks on the analyzer, making necessary adjustments to the instruments, and performing routine maintenance. In addition, a visual inspection of the monitoring site should be made in order to observe the physical condition of the site. Extreme weather conditions or vandals may have caused damage to the site since the last visit.

All operational activities performed on-site should be entered into a logbook as well as onto the strip chart. The logbook is to remain on-site for the duration of the monitoring project.

Shelter Environment

The shelter's role in quality assurance is to provide a temperature controlled environment in which the sampling equipment can operate at its optimum. The shelter temperature must be maintained at 20°C - 30°C (68°F - 86°F) and recorded either real-time on the data logger, or during the site visits in the logbook. For Ecology sites a temperature sensor must be installed at the shelter to record the temperature on a real time basis. Fluctuations outside of these limits should be identified, and the problem corrected.

Figure 5-3
Air Monitor Calibration Report

Location 3rd & Chestnut Date 5/16/93 Operator Jim Werner

Calibrator Model Metronics 450 Serial Number M862 State ID B62637

Analyzer Model Teco 43 Serial Number 7653 State ID B59980

Permeation Tube Serial Number 58108 Permeation Rate 2060 ng/min

Full Scale Operating Range 1.0 PPM

P_a 758 mm HG T_a 22 °C 295 °K

Chamber Temperature 30 °C

Final Zero setting 438 Final Span Setting 716

Analyzer Calibration Relationship: M = 1.013 B = .0020 R = 1.000

Test Point	Recorder Value (%)	(Y) Indicated Concentration (PPM)	(X) Actual Concentration (PPM)
1	0	0	0
2	20	.211	.205
3	40	.408	.399
4	80	.816	.805

$$F = (Pr/PPM) \times (M_c/M_w)$$

F = Flow (cc/min)

Pr = Perm tube perm rate (ng/min)

PPM = Gas concentration desired (ppm)

M_c = Molar constant (22.4 l/mole at 30°C)

M_w = Mole weight of SO₂ (64 g/mole)

Strip Chart Recorder Check

During each weekly visit to the monitoring site, check the following:

- 1) Pen life
- 2) Chart paper for supply
- 3) Chart speed control setting
- 4) Signal input range switch
- 5) Time synchronization (record on chart)

Problems should be identified and corrected immediately. Document all changes in the logbook.

Analyzer Check

Inspect the external operating parameters of the analyzer. These will vary from instrument to instrument, but in general they will include the following:

- 1) Correct settings of flow meters and regulators
- 2) Cycling of temperature control indicators
- 3) Verification that the analyzer is in the sampling mode rather than the zero or the calibration mode.
- 4) Zero and span potentiometers locked and set at proper values.

Calibration Check

Continuous ambient air analyzers require periodic checks to assess the collected data for precision. The calibration check is performed weekly and consists of zero, low, and high span check.

Do not make any adjustments to the analyzer prior to or during a weekly calibration check. The purpose of the calibration check is to evaluate the analyzer performance. Therefore, adjustments made during the calibration check will result in the invalidation of the results. In addition, weekly calibration checks should not be performed during periods of data collection which exceed National Ambient Air Quality Standards (NAAQS) but should be postponed and performed immediately after the air pollution episode.

Allow the analyzer to sample zero air until a stable response is obtained. Record this information on the logbook as well as the strip chart. For the next two calibration check points refer back to the procedures on calibration for instructions on how to generate the required levels of gas concentrations. For the low calibration check point generate a concentration of gas between 0.08 and 0.10 ppm. Allow this point to stabilize. Calculate and record the expected concentration as well as the indicated concentration in the logbook, and on the strip chart. For the high calibration check point, generate a concentration of gas that is approximately 80% of full scale of the

instrument. Again calculate the expected concentration and record this along with the indicated concentration in the logbook and on the strip chart.

The low calibration check point, also called the precision check point should also be entered onto the Washington Department of Ecology Air Monitoring Precision Check Form contained in Section 12. A completed example of this form is shown in Figure 6-1.

If after completing the calibration checks it is found that the instrument does not meet the control limits shown below, then corrective action may be taken (instrument adjustment). These actions should be taken immediately following the completion of the calibration check when results are outside these limits. Do not make any adjustments during the calibration check. In addition, no linear regression calculations are needed during a calibration check.

Quality Control Checks

<u>Check</u>	<u>Control Limit</u>	<u>Corrective Action</u>
Automatic	$> \pm 5\%$	Adjust Analyzer
Daily Low & High Calibration Checks	$> \pm 10\%$	Invalidate Data
Manual Weekly	$> \pm 5\%$	Adjust Analyzer
Low & High Calibration Checks	$> \pm 10\%$	Invalidate Data
Automatic	$> \pm 1\%$ of Full Scale	Adjust Analyzer
Daily & Manual Weekly Zero	$> \pm 2\%$ of Full Scale	Invalidate Data

If adjustment of the analyzer is required, it may be necessary to recheck the zero and span adjustments several times to ensure the analyzer is operating within the control limits.

In addition to the required manual zero, precision, and high calibration checks generated weekly by the operator, automatic calibration checks are triggered daily by the on-site data logger. See the data logger instruction manual for specific instructions on how to set up the automatic calibration checks.

Figure 6-1
Washington Department of Ecology
Air Monitoring Precision Checks

STATION#

1	4	1	4	0	0	1	A
---	---	---	---	---	---	---	---

 YEAR/MONTH: May 1993

ANALYZER STATE TAG #:

B	5	9	9	8	0		
---	---	---	---	---	---	--	--

OPERATOR: Jim Werner POLLUTANT: SO₂

LOCATION: Cosmopolis, School Bus Garage

DATE			ACTUAL CONC				INDICATED CONC				UNITS	PASSED Y or N	COMMENTS
Month	Day	Year											
05	04	93	0.	1	0	0	0.	0	9	6	PPM	Y	
05	10	93	0.	1	0	0	0.	0	9	5	PPM	Y	
05	18	93	0.	1	0	0	0.	0	9	5	PPM	Y	
05	28	93	0.	1	0	0	0.	0	9	8	PPM	Y	

PRECISION CHECK EQUIPMENT:

Cylinder Gas S/N:
 Calibrator Model: Metronics 450
 Calibrator Serial #: M470
 Permeation Tube #: 30-50251

 Shaded areas to be completed by QA Personnel.

7.0 Maintenance

Part of the routine duties include instrument maintenance. Refer to the manufacturer's manual for specific procedures on performing preventive and remedial maintenance.

SO₂ Calibrator Preventive and Remedial Maintenance

To maintain proper operation of the SO₂ calibrator, the following maintenance procedures should be performed periodically;

- 1) Remove the screen from the back panel and clean the filter. If possible, flush the filter with warm water and let dry. Spray the filter with rotron filter coat oil. Clean at least monthly and more often if needed.
- 2) Replace the inline filter at least monthly and more often if operating in a dirty environment.
- 3) Approximately every six months remove the capillary and clean or replace if particulate deposits are present. Replace "O" ring if cut or abraded.
- 4) Check the instrument for pump noise and/or vibrations. Check and/or replace the pump diaphragm every six months or as needed.
- 5) Leak check the instrument by placing the switch in the sample position and plugging the sample inlet with the end of a finger or a cap. If there are no leaks, the rotameter should go to zero.

SO₂ Analyzer Preventive and Remedial Maintenance

The following represents the maintenance that would be performed on a Thermo Electron Corp (TECO) model 43 SO₂ analyzer:

- 1) Approximately every six months remove the capillary and clean or replace if particulate deposits are present. Replace "O" ring if cut or abraded.
- 2) If a sample inline prefilter is used, it should be inspected and replaced monthly, or more often if needed.

A schedule for both the SO₂ analyzer as well as the calibrator should be developed to aid in the completion of the required maintenance tasks.

An example of a preventive maintenance schedule is shown in Figure 7-1. Figures 7-2 and 7-3 are summaries of the suggested maintenance activities and frequency. Any additional information required to perform these maintenance activities can be obtained in the manufacturer's manual. A detailed description of all maintenance activities should be recorded in the on-site logbook. In addition, an example of a weekly station checklist is shown in Figure 7-4. This checklist can be used in conjunction with the onsite logbook to aid in the documentation of onsite maintenance and operational activities.

TECO 43 (SO₂ Analyzer) Preventive Maintenance Schedule 199__

Figure 7-1

PREVENTIVE MAINTENANCE	Manual Refer.	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
Check all internal and external tubing and fittings for leaks, etc.		*	*	*	*	*	*	*	*	*	*	*	*
If chamber pressure increases, inspect and clean or replace capillary. Pressure must = 10"	V-A Page V-1 Fig. V-1 Page V-5				*						*		
Clean fan filter.		*	*	*	*	*	*	*	*	*	*	*	*
Replace sample inlet filter. Check whenever span value drops off. Check MORE OFTEN in dirty area.		*	*	*	*	*	*	*	*	*	*	*	*
Check and/or replace pump diaphragm every six months or as needed.		*						*					
Check output voltages. Step 8 in procedure.		*	*	*	*	*	*	*	*	*	*	*	*

STATION: _____

STATION #: _____

OPERATOR: _____

ANALYZER S/N: _____ STATE TAG #: _____ FROM: _____ TO: _____

Figure 7-2
Routine Calibrator Maintenance Activities

Equipment	Frequency And/Or Method	Acceptance Limits	Action if Requirements are not met.
Particulate filter	Monthly or more often when needed	No obvious particulate deposits	Clean or replace if damaged
Inline 5 micron filter	Monthly or more often when needed	No obvious particulate deposits	Replace as necessary
Capillary	Every six months or more often when dirty	No particulate deposits	Replace both capillary and o-ring if dirty or damaged
Pump	Every six months	No pump noise or vibrations	Replace the pump diaphragm
Leak check calibrator	After any repairs are performed	No leaks	Check all connections for leaks
Tubing and fittings	During weekly visits	No crimps, cracks or obstructions; no cross-threading	Replace as necessary

Figure 7-3
Routine SO₂ Analyzer Maintenance Activities

Equipment	Frequency And/Or Method	Acceptance Limits	Action if Requirements are not met
Inline 5 micron filter	Monthly or more often when needed	No obvious particulate deposits	Replace as necessary
Capillary	Every six months or more often when dirty	No particulate deposits	Replace both capillary and o-ring if dirty or damaged
Tubing and fittings	During weekly visits	No crimps, cracks or obstructions; no cross-threading	Replace as necessary

Figure 7-4
SO₂ Monitor Weekly Station Checklist

SITE: <u>Cosmopolis, Bus Garage</u>		OPERATOR: <u>J. Werner</u>	
	CHECKED	CHECKED	
Pacific Standard Time (PST) entered	<u>11/8</u>		
Calendar date	<u>5/7</u>		
Station Calibrator VICI Metronics 450 State Identification Number <u>B62637</u>			
Power on? (Y/N)	<u>Y</u>		
Is permeation chamber temp at 30.0°C? (Y/N)	<u>Y</u>		
Operating switch in "operate" mode? (Y/N)	<u>Y</u>		
SO ₂ channel disabled on data logger? (Y/N)	<u>Y</u>		
Noted on strip chart? (Y/N)	<u>Y</u>		
Performed calibration check (Y/N)	<u>Y</u>		
Do the recorder and data logger read the same value? (Y/N)	<u>Y</u>		
Dilution flow adjusted following calibration check? (Y/N)	<u>Y</u>		
Operating switch in "remote" mode upon completion of the calibration check? (Y/N)	<u>Y</u>		
SO ₂ channel enabled on data logger? (Y/N)	<u>Y</u>		
Station Analyzer Thermo Electron 43 State Identification Number <u>B59980</u>			
Power on? (Y/N)	<u>Y</u>		
Is the "RANGE PPM" switch in the 0.5 PPM position? (Y/N)	<u>0-1 PPM</u>		
Instrument zero potentiometer setting?	<u>528</u>		
Instrument span potentiometer setting?	<u>716</u>		
Date of last inline filter change?	<u>5/1</u>		
Shelter			
Is the shelter temperature being maintained between 68°F and 82°F? (Y/N)	<u>Y</u>		
Strip Chart Recorder			
Is the time correct on the chart? (Y/N)	<u>Y</u>		
Do the pens need to be changed? (Y/N)	<u>N</u>		
Does the chart paper need replacing? (Y/N)	<u>N</u>		
Is the calibration current? (Y/N)	<u>Y</u>		

8.0 Quality Assurance

Performance Audits

During each calendar quarter, utilizing the procedures and calculations specified in 40 CFR 58, Appendix A, "Quality Assurance Requirements for State and Local Air Monitoring Stations (SLAMS)," at least 50% of the operating automated analyzers will be audited.

System Audits

The systems audit is an on-site review and inspection of the entire ambient air monitoring program to assess its compliance with established regulations governing the collection, analysis, validation, and reporting of ambient air quality data. A systems audit will be performed annually. To provide uniformity in the evaluation, the criteria and procedures specified in EPA's Quality Assurance Handbook for Air Pollution Measurement Systems, Volume II, Section 2.0.11 will be applied.

9.0 Data Recording, Validating and Reporting

Refer to "State of Washington Department of Ecology, Automated Method Data Documentation and Validation Procedures" document. This document describes the procedures for handling, recording and validating air monitoring data. All data will be reviewed and certified by the Quality Assurance Coordinator prior to being reported or used to make decisions concerning air quality, air pollution abatement or control.

10.0 Data Quality Assessment

For each calendar quarter and year, Quality Assurance will prepare data precision, accuracy and completeness reports.

Precision

The precision will be evaluated and reported employing the frequencies, procedures and calculations in 40 CFR 58, Appendix A, "Quality Assurance Requirements for State and Local Air Monitoring Stations (SLAMS)."

Accuracy

Using results from the performance audits and the calculations specified in 40 CFR 58, Appendix A, "Quality Assurance Requirements for State and Local Air Monitoring Stations (SLAMS)," the accuracy will be evaluated and reported.

Data Completeness

The completeness of the data will be determined for each analyzer and expressed as a percentage. Percent valid data will be a gauge of the amount of valid data obtained compared to the amount expected under ideal conditions (24 hours per day, 365 days per year). Exceptions will be made for analyzers which have a seasonal sampling period.

11.0 References

- 1) "Quality Assurance Handbook for Air Pollution Measurement Systems, Volume I - Principles." EPA-600/9-76-005. Dec. 1984.
- 2) "Quality Assurance Handbook for Air Pollution Measurement Systems, Volume II - Ambient Air Specific Methods." EPA-600/4-77-027a. Dec. 1986.
- 3) Code of Federal Regulations, Title 40, Part 58 (40 CFR 58)
- 4) "State of Washington Department of Ecology, Air Quality Program, Automated Method Data Documentation and Validation Procedure." Dec. 1993.
- 5) "State of Washington Department of Ecology, Air Quality Program, Site Selection and Installation Procedures." Oct. 1994.

12.0 Forms

This section contains blank forms. Please make copies as needed retaining the originals within the document.

- SO₂ Siting Criteria Checklist
- Air Monitor Calibration Report
- Air Monitoring Precision Check
- Preventive Maintenance Schedule
- Station Checklist

**Sulfur Dioxide (SO₂)
Siting Criteria Checklist**

Site Name: _____

Site #:

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Address: _____

City: _____ County: _____

1. Vertical distance from ground to probe (3-15 meters): (Probe height must be less than 80% of the mean height of buildings in the neighborhood. This applies whether probe is located on a rooftop or at an intermediate height on a building. The 80% figure applies only to surrounding structures not the building where the probe is located. Minimum height must still be greater than or equal to 3 meters. *NOTE:* Inlet probe must be located away from dirty, dusty areas.) _____ Meters

2. Distance from support structure (> 1 meter): (This separation distance is in reference to walls, parapets or penthouses located on the roof.) Vertical _____ Meters
Horizontal _____ Meters

3. Unrestricted air flow in an arc of 270° or 180° if on side of building. Predominant wind direction must be included in 270° arc. (The inlet probe should be on the windward side of the building relative to the prevailing winter wind direction. At least one of the directions must be the dominant direction for the season of greatest potential pollutant concentration.) Yes _____ No _____

4. Distance from probe to trees (should be > 20 meters). _____ Meters

5. Distance from sampler to any obstacle must be at least twice the height the obstacle protrudes above the sampler. _____ Meters

6. Are any furnace or incineration flues nearby? (The separation distance is dependent on the height of the flue, type of material burned and quality of fuel. Objective is to avoid undue influences from minor local sources.) Yes _____ No _____

7. Comments: _____

Completed by: _____ Date: _____

Agency: _____

Air Monitor Calibration Report

Location_____ Date_____ Operator_____

Calibrator Model_____ Serial Number_____ State ID_____

Analyzer Model_____ Serial Number_____ State ID_____

Permeation Tube Serial Number_____ Permeation Rate_____

Full Scale Operating Range_____ PPM

P_a_____ mm HG T_a_____ °C _____ °K

Chamber Temperature_____ °C

Final Zero setting_____ Final Span Setting_____

Analyzer Calibration Relationship: M = _____ B = _____ R = _____

Test Point	Recorder Value (%)	(Y) Indicated Concentration (PPM)	(X) Actual Concentration (PPM)

$$F = (Pr/PPM) \times (M_c/M_w)$$

F = Flow (cc/min)

Pr = Perm tube perm rate (ng/min)

PPM = Gas concentration desired (ppm)

M_c = Molar constant (22.4 l/mole at 30°C)

M_w = Mole weight of SO₂ (64 g/mole)

STATION#

YEAR/MONTH: _____

--	--	--	--	--	--	--	--

POLLUTANT: _____

[illegible]

Permeation Tube #: _____

Decimal Placement		
CO	1	XX.X ppm
SO ₂	3	.XXX ppm
Ozone	3	.XXX ppm
Neph	2	X.XX Bscat

 Shaded areas to be completed by QA Personnel.

TECO 43 (SO₂ Analyzer) Preventive Maintenance Schedule 199__

PREVENTIVE MAINTENANCE	Manual Refer.	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
Check all internal and external tubing and fittings for leaks, etc.		*	*	*	*	*	*	*	*	*	*	*	*
If chamber pressure increases, inspect and clean or replace capillary. Pressure must = 10"	V-A Page V-1 Fig. V-1 Page V-5				*						*		
Clean fan filter.		*	*	*	*	*	*	*	*	*	*	*	*
Replace sample inlet filter. Check whenever span value drops off. Check MORE OFTEN in dirty area.		*	*	*	*	*	*	*	*	*	*	*	*
Check and/or replace pump diaphragm every six months or as needed.		*						*					
Check output voltages. Step 8 in procedure.		*	*	*	*	*	*	*	*	*	*	*	*

STATION: _____

STATION #: _____

OPERATOR: _____

ANALYZER S/N: _____ STATE TAG #: _____ FROM: _____ TO: _____

SO₂ Monitor Weekly Station Checklist

SITE: _____ OPERATOR: _____		
	CHECKED	CHECKED
Pacific Standard Time (PST) entered		
Calendar date		
Station Calibrator VICI Metronics 450 State Identification Number _____		
Power on? (Y/N)		
Is permeation chamber temp at 30.0°C? (Y/N)		
Operating switch in "operate" mode? (Y/N)		
SO ₂ channel disabled on data logger? (Y/N)		
Noted on strip chart? (Y/N)		
Performed calibration check (Y/N)		
Do the recorder and data logger read the same value? (Y/N)		
Dilution flow adjusted following calibration check? (Y/N)		
Operating switch in "remote" mode upon completion of the calibration check? (Y/N)		
SO ₂ channel enabled on data logger? (Y/N)		
Station Analyzer Thermo Electron 43 State Identification Number _____		
Power on? (Y/N)		
Is the "RANGE PPM" switch in the 0.5 PPM position? (Y/N)		
Instrument zero potentiometer setting?		
Instrument span potentiometer setting?		
Date of last inline filter change?		
Shelter		
Is the shelter temperature being maintained between 68°F and 82°F? (Y/N)		
Strip Chart Recorder		
Is the time correct on the chart? (Y/N)		
Do the pens need to be changed? (Y/N)		
Does the chart paper need replacing? (Y/N)		
Is the calibration current? (Y/N)		